

A screened Rydberg formula for the successive ionization energies of all atoms and ions

Muhammad Awais

Independent researcher, Multan, Pakistan. Correspondence: mawais9171@gmail.com

****Abstract—**We present a closed-form expression, the *screened Rydberg formula*, for the successive ionization energy $IE(Z, N)$ of any atom or ion. Its inputs are the nuclear charge Z and the ground electron configuration. The formula keeps the hydrogenic form $Ry Z_{\text{eff}}^2/n^2$ and splits the effective charge into two parts. The first is the first-order screening constant $\sigma_1 = -n^2 \Delta E_1$ of the $1/Z$ perturbation expansion. It has no adjustable parameter; it is a rational number for every configuration, and we tabulate it for $N = 1-110$. The second is a fitted higher-order remainder that depends on the ion charge in an Edlén-type form. A saturation construction keeps $Z - N + 1 \leq Z_{\text{eff}} \leq Z$ for any parameter values (given $0 \leq \sigma_1 \leq N - 1$, which holds for every configuration we use) and preserves the exact Z^2 and Z coefficients of the $1/Z$ series. One-electron Dirac, recoil, finite-nuclear-size and QED corrections enter without fitted parameters. The final model has 33 global parameters and none per element or ion. On the 5847 successive ionization energies in the NIST Atomic Spectra Database ($Z = 1-110$) its mean absolute percentage error (MAPE) is 1.87 % (median 0.94 %). It is 7.52 % for the first ionization energies of neutral atoms and 0.0012 % for hydrogen-like ions; the H-like figure agrees with, but is not independent of, the QED theory behind the NIST values. We chose the model with a pre-registered score that never used the extrapolation test, although its interpolation splits contain heavy elements. Fitted on $Z \leq 54$ and used to predict $Z = 55-110$, the model gives 6.60 % MAPE (median 1.62 %). On the same rows Slater's rules give 11.6 %, a published parameter-free screened-hydrogenic model 13.6 %, and two earlier fitted variants without the bound 57.2 % and 170 % (Table 1). Heavy neutral atoms remain the weak point, with 21.8 % MAPE on their first ionization energies in that test. The $Z \leq 54$ fit puts Pb and Tl at about 1.2 and 1.7 eV (NIST 7.42 and 6.11 eV) and makes Lr negative, and the all-data fit puts oganesson (3.66 eV) far below radon (NIST 10.75 eV). We report these failures and have not corrected them.

Index Terms— ionization energy; screening constants; $1/Z$ expansion; screened hydrogenic model; isoelectronic sequences; Slater's rules; blind validation.

1. Introduction

The Bohr energy $Ry Z^2/n^2$ is exact only for a non-relativistic one-electron ion. For every other atom and ion the ionization energy (IE) depends on electron–electron repulsion, which has no closed-form solution for $N \geq 2$. Four lines of work bear on the problem.

Slater [1] replaced Z by an effective charge $Z - \sigma$, with σ given by counting rules fitted by hand to atomic data. Clementi and Raimondi [2], [3] obtained σ from optimised self-consistent-field orbital exponents of neutral atoms. Both were designed for orbitals of neutral atoms, not for the full set of successive IEs. Scored as IE formulas on all NIST ions with a total-energy difference, Slater's rules give 11.8 % MAPE and Clementi–Raimondi 120 % (§4).

Layzer [4], [5] showed that the non-relativistic energy of a fixed configuration is an asymptotic series $E = Z^2 E_0 + Z E_1 + E_2 + \dots$. E_0 is hydrogenic, and E_1 is a rational combination of hydrogenic Slater integrals, which fixes the $Z \rightarrow \infty$ limit of the screening constant exactly. Higher orders need continuum sums [6], [7]. Along isoelectronic sequences, Edlén [8] described the smooth variation of screening with ion charge by expansions in $1/(\zeta + s)$. Z -expansion codes with relativistic corrections were developed for selected sequences [9].

Screened hydrogenic models (SHMs), starting with Mayer [10] and More [11], are used in plasma-physics codes. Their screening constants have been tabulated or fitted with l -splitting [12], [13], derived from analytical potentials [14], [15], fitted by a genetic algorithm with relativistic subshells [16], or computed self-consistently as functions of Z and N without parameters [17]–[22]. Average-atom and kinetics codes use SHMs because they are fast and cover all ions [23]–[25].

Ab initio methods give more accurate numbers: Koopmans' theorem [26], Δ SCF Hartree–Fock and Kohn–Sham DFT [27], [28], correlated non-relativistic energies [29], and Dirac–Fock total energies for all ground configurations up to $Z = 118$ [30]. They are numerical procedures, not formulas.

We found no closed-form expression that covers every ion of every element from the configuration alone, reproduces the large- Z behaviour of the $1/Z$ expansion exactly, reports every fitted parameter, and is validated on all NIST successive IEs with held-out tests fixed in advance, including an extrapolation to heavier elements. The SHM papers we read report accuracy on their fitting data or, like [20], on selected sequences. Scored on our rows (§4.5), the constants of Mendoza et al. give 2.82 % MAPE on the 5011 rows they cover. This paper addresses that gap. It does not propose a new law of atomic physics, and §5.1 lists what is rediscovered.

The paper makes four contributions. It tabulates σ_1 for every NIST ground configuration with $N = 1$ –110, as a parameter-free counterpart to Slater's σ that is exact as $Z \rightarrow \infty$ (§2.1, §4.6). It defines the screened Rydberg formula, σ_1 plus a bounded Edlén-type remainder with 33 global parameters (§2.2–2.4). It applies one validation protocol, with a pre-registered selection score and a single extrapolation test, to the final model, its ancestors and the published baselines (§3, §4). And it compares the formula with two published SHMs, re-implemented and scored with the same scorer (§4.5): the parameter-free Kregar/Di Rocco model on all 5847 rows, and the fitted constants of Mendoza et al. (2011) on the 5011 rows they cover.

2. Theory

§2.1 uses atomic units; energies are converted to eV with $Ry = 13.6057$ eV. Z is the nuclear charge, N the number of electrons before ionization, and $Z_a = Z - N + 1$ the charge seen by the departing electron. (n, l) is the removed subshell and k its occupancy.

2.1 Exact first-order screening from the $1/Z$ expansion

Scaling $r \rightarrow \rho/Z$ turns the non-relativistic Hamiltonian into $H = Z^2[H_0 + Z^{-1}V]$, with H_0 hydrogenic and $V = \sum 1/\rho_{ij}$. Rayleigh–Schrödinger perturbation theory in $1/Z$ then gives, for a fixed configuration and term, $E(Z, N) = Z^2E_0 + ZE_1 + E_2 + \dots$, and the ionization energy is

$$IE(Z, N) = Z^2\Delta E_0 + Z\Delta E_1 + \Delta E_2 + \dots, \quad \Delta E_k = E_k(N-1) - E_k(N).$$

At zeroth order $\Delta E_0 = 1/(2n^2)$, which is Bohr's formula. At first order $E_1 = \langle V \rangle$ over the $Z = 1$ hydrogenic state. Every hydrogenic radial integral F^k and G^k is a rational number, because each product $P_a P_c$ is a polynomial times $e^{-\beta r}$ with rational β ; for example $F^0(1s, 1s) = 5/8$, $F^0(1s, 2s) = 17/81$ and $G^0(1s, 2s) = 16/729$. We evaluate the integrals in exact rational arithmetic and combine them with exact angular algebra. For open subshells E_1 is evaluated for the Hund ground term by projecting onto the highest-weight (S, L) subspace. Where hydrogenic degeneracy couples configurations (the Layzer complex, e.g. $1s^2 2s^2$ with $1s^2 2p^2$), V is diagonalised within the complex. For two or more open subshells we use a high-spin coupled term, which is the one approximation in σ_1 .

Completing the square in the first two terms defines the first-order screening constant

$$IE \simeq \Delta E_0 (Z - \sigma_1)^2, \quad \sigma_1 = -\frac{\Delta E_1}{2\Delta E_0} = -n^2 \Delta E_1.$$

σ_1 is Layzer's $Z \rightarrow \infty$ screening constant. It has no adjustable parameter and depends on the configuration, not on Z . For He-like ions $E_1(1s^2) = 5/8$, so $\sigma_1(\text{He}) = 0.6250$ (Slater: 0.30). For Li-like ions $E_1(1s^2 2s) = 5965/5832 = 1.0228052 = 5/8 + 2 \cdot 17/81 - 16/729$, so $\Delta E_1 = -0.397805$ and $\sigma_1 = 1.5912$. For Be-like ions the complex value $E_1 = 1.5592742$ agrees with Layzer's.

The series fails for near-neutral atoms (§4.6), because its expansion parameter is effectively N/Z . For a neutral atom $Z - \sigma$ is only 1–3, so a 10 % error in σ becomes a 100–1000 % error in IE. The higher orders have no closed form: E_2 already requires sums over the hydrogenic continuum. The completed-square guess $\Delta E_2 = \Delta E_1^2/(4\Delta E_0)$, for example, is 24–26 % too large for He- and Li-like ions. For this reason the remainder in §2.2 is fitted.

2.2 The screened Rydberg form with a bounded remainder

The final model (code name `pa_hier_rel`) is

$$\text{IE}(Z, N) = \mu(Z) \left\{ \text{Ry} \frac{Z_{\text{eff}}^2}{n^2} F_{n,j}(Z_{\text{eff}}) \left[1 + r_c \frac{(Z\alpha)^2}{n} \left(\frac{Z_{\text{eff}}}{Z_a} - 1 \right) \right] + \text{Ry} \frac{x_l K_l(k)}{n^2} \right\} - [\Delta E_{\text{QED}} + \Delta E_{\text{FNS}}]_{Z,n} \left(\frac{Z_{\text{eff}}}{Z} \right)^2,$$

where the last term applies only to the removal of an ns electron with $n \leq 2$, and

$$Z_{\text{eff}} = Z - \sigma_1(\mathcal{C}) - D, \quad D = \frac{T}{Z_a + \kappa + |T|/\hbar}, \quad h = \begin{cases} (N-1) - \sigma_1 & T \geq 0 \\ \sigma_1 & T < 0 \end{cases}, \quad T = \sum_g \tau_g \nu_g + \sum_c \delta\tau_c \nu_c.$$

Each of the other $N - 1$ electrons, in a subshell (n', l') , is counted in exactly one of five screening groups ν_g . The group *same* holds the $k - 1$ other electrons of (n, l) . The group *out* holds outer electrons, with $n' > n$ or with $n' = n$ and $l' > l$. All remaining electrons are inner. For an s or p target, inner electrons with $n' \geq n - 1$ form *in* (the ns electrons of an np target plus the whole $(n - 1)$ shell, including its d and f electrons) and those with $n' \leq n - 2$ form *core*. For a d or f target, every inner electron, from the same- n lower- l subshells down to 1s, is in *df*. Hence $\sum_g \nu_g = N - 1$, and each group has a coefficient τ_g . As $(\nu_{\text{same}}, \nu_{\text{in}}, \nu_{\text{core}}, \nu_{\text{df}}, \nu_{\text{out}})$, 0 $2p^4$ gives (3, 4, 0, 0, 0) and Na 3s gives (0, 8, 2, 0, 0). Fe $3d^6 4s^2$ with 4s removed gives (1, 14, 10, 0, 0), because $3d^6$ belongs to the $(n - 1)$ shell. Pb $6p^2$ gives (1, 20, 60, 0, 0), with $6s^2 5s^2 5p^6 5d^{10}$ in *in* and every electron with $n' \leq 4$, including $4f^{14}$, in *core*.

The screening classes ν_c (Table A2) refine the groups. An electron's class depends on the target type (s/p, d or f), on $n - n'$ and on l' . Each class lies inside one group, and an electron of class c in group g contributes $\tau_g + \delta\tau_c$ to T . A ridge penalty (10^{-4} per row) shrinks the deviations toward their group value. Two of the 21 classes are empty in all 5847 rows and carry no $\delta\tau_c$: *sn_out* ($n' = n, l' > l$) and *out_sp* ($n' > n$, s/p target). That leaves the 19 of Table A1. The bounded 9-parameter variant (`pa_bound9`) and the pocket formula (Appendix A) use the five groups without classes. An independent implementation of the rule reproduces the code's group and class counts with 0 mismatches on all 5847 rows and on all 7021 configurations with $Z \leq 118$ (`tools/check_grouping_rule.py`).

The removed subshell (n, l) is the one whose occupancy drops from the N -electron configuration to the NIST ground configuration of the $(N - 1)$ -electron ion. The N -electron configuration is the NIST ground configuration, or the Madelung order if the ion is not tabulated. For a rearranging ion (e.g. V $3d^3 4s^2 \rightarrow V^+ 3d^4$) it is the subshell that loses most electrons, with ties going to larger n and then larger l . If the $(N - 1)$ -electron ion is not tabulated, or the user supplies the configuration, the outermost subshell (largest n , then largest l) is removed.

The Hund term uses $K_l(k) = [P(k) - P(k-1)] - 2l(k-1)/(4l+1)$, the change in the number of parallel-spin pairs relative to a statistical average. $P(k)$ is the number of parallel-spin pairs of l^k under Hund's first rule. Equivalently $K_l(k) = (2l+1)(k-1)/(4l+1)$ for $k \leq 2l+1$ and $K_l(k) = -K_l(4l+3-k)$ above half filling, so $K_s \equiv 0$, $K_p = 0, 0.6, 1.2, -1.2, -0.6, 0$ for $k = 1 \dots 6$, $K_d = (5/9)(0, 1, 2, 3, 4, -4, -3, -2, -1, 0)$ and $K_f = (7/13)(0, 1, \dots, 6, -6, \dots, -1, 0)$. The amplitude x_l is fitted for $l = p, d, f$; s electrons have no Hund term.

Three inputs need a precise definition. σ_1 is evaluated for the *frozen* configuration, the N -electron configuration minus one (n, l) electron, and not for the ground configuration of the ion. The two differ only for the 63 rearranged rows, and the choice keeps $\Delta E_0 = 1/(2n^2)$ on those rows. The reduced-mass factor is $\mu(Z) = M/(M + m_e)$, where M is the nuclear mass of the isotope used in the QED tabulation [31]. The QED and finite-nuclear-size shift is applied only to 1s and 2s removal (427 rows); it is the one-electron shift of §2.4 for charge Z , scaled by $(Z_{\text{eff}}/Z)^2$, and reaches at most 0.9 % of the IE (at $Z = 110, N = 2$). For $N = 1, T = 0$ and so $D = 0$.

A reader cannot reconstruct the frozen-configuration σ_1 , $\mu(Z)$ or the QED/FNS shift from the text alone, so we publish every per-row input in `results/model_inputs.csv`: removed subshell, $k, j, \sigma_1, \nu_g, \nu_c, K, \mu$ and QED/FNS. A short script that uses only this table, the equations above and the parameters of Table A1 (`tools/verify_from_inputs.py`, which imports no model code) reproduces the production code on all 5847 rows to $7 \cdot 10^{-16}$ relative, for the final model and for `pa_bound9`.

The saturation form of D has two properties for any parameter values. First, it bounds the effective charge, $Z_a \leq Z_{\text{eff}} \leq Z$. Writing $D = \text{sign}(T) \cdot h \cdot u / (1 + u)$ with $u = |T|/[h(Z_a + \kappa)]$ gives $|D| < h$, so Z_{eff} can fall neither below the fully screened, non-penetrating limit Z_a nor above the bare charge Z . Because $h = (N-1) - \sigma_1$ or σ_1 , this requires $0 \leq \sigma_1 \leq N - 1$. We have checked rather than proved that condition: it holds for all 7021 ground and Madelung configurations with $Z \leq 118$, and the code warns if a user-supplied configuration breaks it. Second, it keeps the exact asymptotics. At fixed N

and $Z_a \rightarrow \infty$, $D \rightarrow T/(Z_a + \kappa) = O(1/Z)$, so the expansion of IE begins $Ry[Z^2 - 2Z\sigma_1 + O(1)]/n^2$ and its Z^2 and Z coefficients are those of the exact $1/Z$ series. The fitted part represents only ΔE_2 and higher orders, that is relaxation, correlation and penetration at low ion charge.

2.3 Charge-dependent penetration as an Edlén-type term

We found the remainder's dependence on $1/(Z_a + \kappa)$ empirically (docs/semi_empirical.md). Along every isoelectronic sequence the excess charge $p = Z_{\text{eff}} - Z_a$ grows with ion charge q as $p \propto -\tau/(Z_a + \kappa)$, with one κ for all sequences. For the Na sequence (3s), $p = 0.84, 1.15, 1.34, 1.46, 1.56$ for $q = 0-4$ and 2.11 at $q = 20$. Adding the term reduced the error of the purely fitted model from about 9 % to about 2 % MAPE. This is an independent rediscovery of the isoelectronic regularity that Edlén formalised [8], and it is consistent with Layzer's theory, in which the screening constant is $\sigma_0 + \sigma_1'/Z + \dots$. What is specific here is narrower: one denominator for all sequences, combined with exact σ_1 and with the saturation bound of §2.2.

2.4 Relativistic, recoil, finite-size and QED corrections

$F_{\{n,j\}}(\zeta)$ is the exact ratio of the point-nucleus Dirac binding energy to the Schrödinger energy for charge ζ . With $x = \zeta\alpha$ and $\kappa = j + \frac{1}{2}$, $F = (2n^2/x^2) \cdot \{1 - [1 + (x/(n - \kappa + \sqrt{\kappa^2 - x^2}))^2]^{-1/2}\}$. j follows jj filling, $l - \frac{1}{2}$ while $k \leq 2l$ and $l + \frac{1}{2}$ otherwise. We write F to avoid confusion with the screening remainder D .

The bracket $1 + r_c(Z\alpha)^2(Z_{\text{eff}}/Z_a - 1)/n$ is a Fermi–Segrè-type correction for the penetration of the valence electron into the region of full nuclear charge. Its five coefficients r_c , for the classes s, $p^{1/2}$, $p^{3/2}$, d and f, are fitted.

For one-electron ions we use the Dirac energy with a numerically solved finite-nuclear-size (FNS) shift, Barker–Glover recoil, the Uehling vacuum polarisation computed over the Dirac 1s density, and the one-loop self-energy function $F_{\text{SE}}(Z\alpha)$ from the all-order tabulation of Yerokhin and Shabaev [31]. The 2×110 tabulated values are external theory inputs, not fitted parameters. In the many-electron formula the QED and FNS shifts are applied only to 1s and 2s removal, scaled by $(Z_{\text{eff}}/Z)^2$. The closed-form low-order $Z\alpha$ expansion of F_{SE} cannot replace the table: it matches the table at $Z = 1$ but diverges for $Z \gtrsim 15$, and it gives 0.354 % MAPE on H-like ions, worse than no QED.

2.5 Density functional theory as a physics check

We also wrote a radial Kohn–Sham LSDA solver (Slater exchange plus VWN5 correlation). It reproduces the NIST LDA reference total energies [28] to about 10^{-6} hartree (e.g. Ne -128.233481). That reference uses the same functional, so the agreement verifies the implementation, not the physics. ΔSCF ionization energies from this solver have no fitted parameters and serve as an independent check that is not a formula (§4.2). DFT does not enter the formula.

2.6 Parameter count

The final model has 33 fitted global parameters:

block	count
group screening τ_g	5
saturation denominator κ	1
class deviations $\delta\tau_c$ (ridge-shrunk; counted in full)	19
relativistic penetration r_c	5
Hund amplitudes x_l (p, d, f)	3

There are no per-element or per-ion parameters, and the prediction code never reads a NIST ionization energy. The fitted values are in Table A1 (Appendix) and `results/uni_final_params.json`; the class definitions are in Table A2.

Each model name in this paper refers to one model:

name	params	definition
Screened Rydberg formula (final; code name <code>pa_hier_rel</code>)	33	eq. (2.2) with groups, classes, relativistic bracket
bounded 9-parameter variant (<code>pa_bound9</code>)	9	eq. (2.2) with the five groups only (no $\delta\tau_c$) and no relativistic bracket ($R = 1$)
pocket formula (<code>uni_pocket</code>)	8	Appendix A. No σ_1 , no bound: $Z_{\text{eff}} = Z - \sum s_g v_g - t(N-1)/(Z_a + \kappa)$

pa_bound9 uses the pocket formula's five electron groups but is a different model.

3. Data and validation protocol

3.1 Data

The reference data are the 5847 successive ionization energies of the NIST Atomic Spectra Database [32] for $Z = 1$ –110, all charge states, with NIST ground configurations (`data/nist_ie.csv`). We took them from ASD version 5.12 [32], the current version at the time, on or before 5 October 2026; the project log records results computed from these data on that date, and the original download timestamp was not kept. The database flags 311 values as experimental, 919 as semi-empirical and 4617 as theoretical. Every metric below uses all 5847 rows unless stated otherwise. By status, the final model's all-data fit gives 4.63 % MAPE on experimental rows (median 2.86 %, $n = 311$), 2.20 % on semi-empirical rows (median 0.83 %, $n = 919$) and 1.62 % on theoretical rows (median 0.92 %, $n = 4617$) (`evaluate.py, status= strata`). The experimental rows are mostly neutral atoms and low-charge ions (98 neutral, 193 of 311 with charge ≤ 2 , median charge 2), the hardest regime for the formula, so their higher error reflects the neutral-atom weakness rather than a disagreement with experiment.

Three subsets recur below: the 108 neutral-atom first IEs, the 110 hydrogen-like ions, and 63 rearranged ions whose ground configuration changes by more than one electron on ionization (e.g. $V\ 3d^34s^2 \rightarrow V^+ 3d^4$).

The configuration of each ion is an input. Inside the table it is the NIST ground configuration. For ions outside it we use the Madelung order: all ions with $Z > 110$, and 258 ions with $Z = 104$ –110 that the table lacks. These ions enter no fit and no score, only the 7021-configuration checks (§4.7) and predictions outside the table.

The error measure is $MAPE = (100/M) \sum |IE_{pred} - IE_{NIST}| / IE_{NIST}$, together with the median absolute percentage error. All scores come from one shared scorer (`evaluate.py`). Fits minimise squared log-ratios $\ln(IE_{pred}/IE_{NIST})$, so a 4 eV and a 100 keV ionization energy carry equal relative weight.

3.2 Held-out splits

Every fitted model is refit on the training rows of each split with its own fitting routine:

split	training rows	test / validation rows	n test rows	role
V1	$Z \leq 36$	$37 \leq Z \leq 54$	819	selection
V2	$Z \leq 44$	$45 \leq Z \leq 54$	495	selection
S1	$Z \bmod 5 \neq 0$	$Z \bmod 5 = 0$ (unseen elements)	1188	selection
S3	$N \bmod 6 \neq 0$	$N \bmod 6 = 0$ (unseen isoelectronic sequences)	928	selection
S2	$Z \leq 54$	$Z \geq 55$	4362	blind , reported once

The selection score is the mean MAPE over V1, V2, S1 and S3. It was fixed before the final round of model development and was the only quantity used to choose the final model. S2 (fit $Z \leq 54$, predict $Z \geq 55$) was never used for a choice: we computed it once per candidate, after the candidate's specification was frozen. The selection splits do contain heavy elements. As pre-registered, S1 and S3 have $Z \geq 55$ rows in both training and test sets (913 of the 1188 S1 test rows and 703 of the 928 S3 test rows, about 76 %), so heavy-atom *interpolation* accuracy informed the design choices and only heavy-atom *extrapolation* stayed blind. As a robustness check, made after the audit and for reporting only, we recomputed the selection score with S1 and S3 restricted to $Z \leq 54$ in training and test. The ranking is unchanged: pa_hier_rel 1.844, the 28-parameter variant without the relativistic term 1.989, the bounded 9-parameter variant 2.094 and the lowest-scoring Push B candidate 2.118. This checks the ranking of the frozen candidates, not the exploration that produced them. An independent validation script reproduced the final model's numbers, matching the developing agent's own run with a difference of 0.0 in every split.

3.3 History of the protocol

The protocol above is the project's third. In an earlier stage a purely fitted model reported S2 = 19 % using a bound on core screening chosen after seeing S2; without that bound its S2 value is 170 %. Later, an intermediate unified model

was chosen on a score that included S2. We report both only as non-blind history (§4.3).

Several other facts bear on how blind the final choice was. The developers of the final round knew which heavy near-neutral atoms the previous model failed on, and that an 8-parameter model extrapolated well; this shaped the direction of the search, though the choice itself was made on the selection score. About 57 variants were explored in that round (43 logged by one agent and about 14 by the other), so the minimum selection score over them is optimistic. The screening classes and the Hund term were designed earlier on all rows, including $Z \geq 55$, which is structural leakage that cannot be removed now.

After the first draft of this paper we tried four more pre-registered variants (`docs/preregistration_v2.md`, `models/v2/NOTES.md`): a relativistic factor that cannot change sign, a second-order charge term, and a κ per orbital type. The sign-preserving relativistic factor fixes the signs of Lr and Og, but no variant improved the selection score (2.15 $\rightarrow$ 2.18–2.49), so the model was left unchanged. That brings the number of explored variants to about 61. The pre-registration file was fingerprinted (`sha256`, `models/v2/PREREG_HASH.txt`) before that round and committed to git afterwards.

3.4 Computational reproducibility

The frozen run used Python 3.13 with NumPy and SciPy; their exact versions were not recorded. We repeated every refit in a second software environment (Python 3.11.9, NumPy 2.4.4, SciPy 1.17.1; `docs/review/reproducibility_py311.md`). The final model reproduces to within 0.005 percentage points in every split: selection score 2.1473 vs 2.1481, blind S2 6.6030 vs 6.6031 %, and 1.8742 % on all data in both. The unbounded reference models depend on the least-squares path. The pocket formula's V1 fit, which diverges in the frozen run (MAPE $4.4 \cdot 10^6$ %), converges in the second environment to 4.35 % (selection score 4.19). The u29 V1 fit converges in the frozen run and diverges in the second, and u35's blind S2 moves from 57.2 % to 56.5 %. Their V1 entries in Table 1 therefore describe the optimizer as much as the model, and we do not use the pocket formula's V1 divergence as evidence against it.

4. Results

4.1 Baselines versus the final model

Every cross-model number in this section comes from one generated matrix with the row count in each cell, `results/benchmark_matrix.md` (`tools/benchmark_matrix.py`), and comparisons between models always use the same rows.

Table 1. MAPE in %. "All", "neutral" and "H-like" are from all-data fits; V1, V2, S1, S3 and S2 are held-out values after refitting. Parameter-free models are not fitted, so their split columns are the error on those rows. Sources: `results/model_comparison.md`, `results/uni_validation.md`.

model	fitted params	all (median)	neutral 1st IE	H-like	V1	V2	S1	S3	selection score	blind S2 (median / neutral)
Bohr Ry Z^2/n^2	0	1363 (172)	$2.3 \cdot 10^4$	6.00	1050	1045	1317	1399	1203	1517
Slater 1930, total-energy difference	0	11.8 (7.48)	51.8	6.00	13.4	14.4	11.6	12.4	12.9	11.6 (8.07 / 50.7)
Clementi–Raimondi, total- energy difference	0	120 (18.7)	2281	6.00	87.1	77.3	116	119	99.9	134
Exact 1/Z series, completed square + rel/QED ("zexp")	0	69.8 (7.20)	1105	0.0012	53.4	54.0	68.3	74.3	62.5	77.8 (– / 1772)
GSHM final (purely fitted, blind)	32	1.57 (0.74)	7.61	0.19	34.1	1.65	1.59	1.70	9.77	170 (3.51 / 7393)
exact σ_1 + linear remainder ("u35")	35	1.48 (0.73)	6.43	0.0012	26.4	1.48	1.52	1.65	7.76	57.2 (2.00 / 3831)
pocket formula (hand- calculable)	8	4.68 (3.04)	16.8	1.16	$4.4 \cdot 10^6$ or 4.35†	2.55	4.62	5.24	– or 4.19†	11.3 (2.04 / 12.3)

model	fitted params	all (median)	neutral 1st IE	H-like	V1	V2	S1	S3	selection score	blind S2 (median / neutral)
bounded 9-parameter variant (pa_bound9)	9	2.90 (1.39)	12.0	0.0012	2.16	1.95	2.90	3.05	2.51	7.80 (– / 28.0)
Screened Rydberg formula (final, "pa_hier_rel")	33	1.87 (0.94)	7.52	0.0012	3.09	1.60	1.91	2.00	2.15	6.60 (1.62 / 21.8)

† Optimizer-path dependent (§3.4): the V1 fit diverges in the frozen run and converges to 4.35 % in a second software environment.

Slater's rules have an all-row MAPE of 11.8 %, 6.3 times that of the final model (1.87 %) and 4.1 times that of pa_bound9 (2.90 %). Their median is 7.48 % against 0.94 %, and their blind S2 error is 11.6 % against 6.60 %, a factor of 1.8 (ratios from results/benchmark_matrix.md).

The final model has the lowest selection score of all candidates (2.15). The next were the other developer's choice, pb_clip_pos (30 parameters), at 2.37, and the 28-parameter hierarchical variant pa_hier at 2.37, with blind S2 errors of 13.9 % and 8.47 %. The bound costs some accuracy inside the data, 1.87 % against 1.48 % for u35 and 7.52 % against 6.43 % on neutral atoms, and in return it removes the catastrophic extrapolation failures.

4.2 Errors by stratum

Table 2. All-data fit, MAPE in % (source: results/lit_comparison.md / docs/literature.md §2.2).

model	N ≤ 10	11 ≤ N ≤ 36	N ≥ 37	Z ≥ 55	removed d	removed f
Slater 1930	4.89	9.38	16.8	11.6	16.2	12.9
pocket formula (8 p)	2.37	2.76	7.38	4.7	3.67	11.5
GSHM final (32 p)	0.698	1.16	2.31	1.48	0.982	2.67
u35 (35 p)	0.463	1.08	2.28	1.58	1.17	2.58
final (33 p)	0.439	1.43	2.87	1.92	1.52	3.51

The error grows with the number of electrons and falls with ion charge; on ions of charge ≥ 3 the final model gives 1.61 %. For few-electron, highly charged ions the parameter-free exact series is already good, with a median of 0.7 % for $N \leq 10$ and 0.48 % MAPE for $N/Z \leq 0.2$ with two terms. On neutral atoms the parameter-free LSDA Δ SCF solver reaches 3.30 % on the 54 first IEs with $Z \leq 54$, and 0.80 % on all 171 ions with $Z \leq 18$. It beats every closed form on neutral atoms, but it is a numerical procedure and was run on only 207 rows.

Further figures are in the repository: a parity plot (results/figures/uni_parity.png), first IEs against Z (results/figures/uni_first_IE.png), successive IEs of selected elements (results/figures/uni_successive.png) and residuals (results/figures/uni_residuals.png).

4.3 Blind extrapolation (S2)

Fitted on $Z \leq 54$ only, the final model predicts the 4362 ionization energies of $Z = 55\text{--}110$ with 6.60 % MAPE and a 1.62 % median. Its ancestors show what each ingredient contributes:

model	what it adds	blind S2 MAPE (%)	S2 neutral (%)
GSHM final (fitted screening, no exact σ_1)	–	170	7393
u35	exact σ_1 as the large-Z anchor, linear remainder	57.2	3831
final	+ saturation bound $Z_a \leq Z_{\text{eff}} \leq Z$, hierarchical shrinkage	6.60	21.8
Slater 1930 (0 p, nothing extrapolated)	–	11.6	50.7

Exact σ_1 alone did not fix extrapolation; confining the fitted part to a bounded remainder did. In a 14-parameter ablation the bound improved the selection score from 3.96 to 2.91, and its blind S2 changed from 9.43 % to 5.53 %, a change we saw only after the choice. For ions ($Z > N$) the blind S2 MAPE is 6.4 %; for neutral atoms it is 21.8 %. Figure 1 shows the per-element median error on the S2 test rows.

Figure 2 plots the selection score against blind S2 for all fitted candidates. Candidates with a low selection score generally extrapolate well, but the relation is not monotone. One candidate with a worse selection score (pb_exp_pos, 3.73) has a lower blind S2 (6.03 %) than the final model. We did not promote it, because that would be selection on S2.

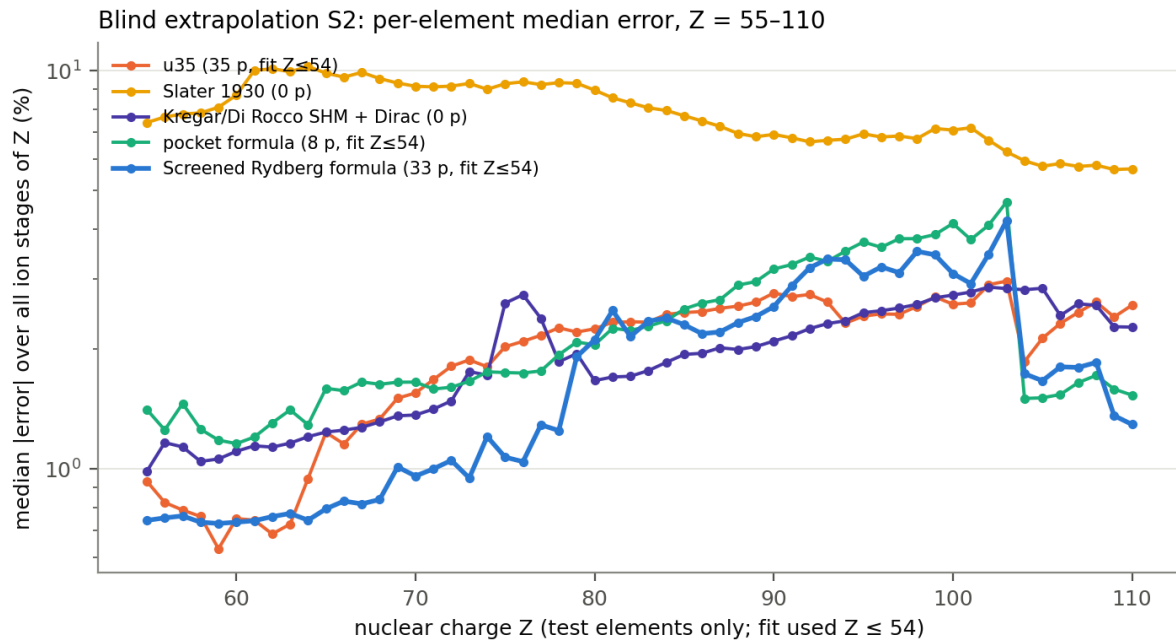


Figure 1. Blind extrapolation. Each point is the median absolute percentage error over all ion stages of one element $Z = 55-110$. The three fitted models are fitted on $Z \leq 54$ only; Slater's rules and the Kregar/Di Rocco SHM have no fitted parameters. The S2 MAPEs, computed from refits with the project's validation code, reproduce Table 1 (final 6.603 %, u35 57.233 %, pocket 11.334 %, Slater 11.638 %).

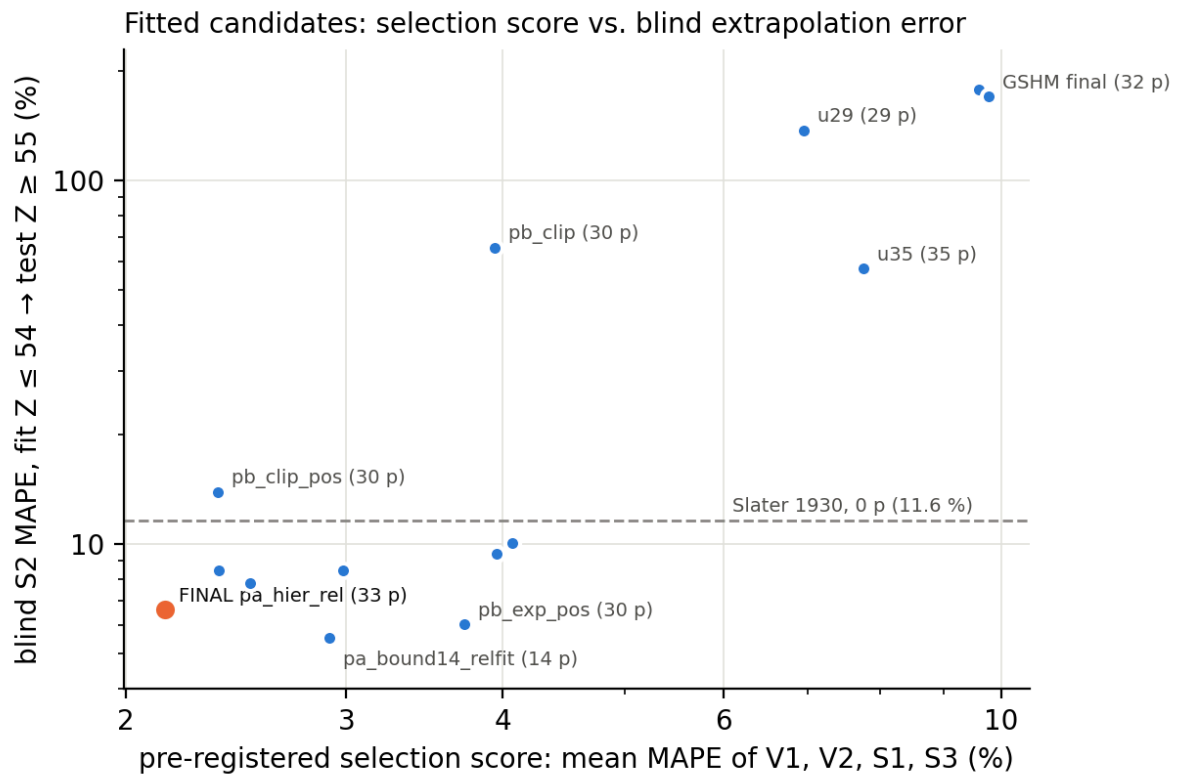


Figure 2. Selection score (mean of V1, V2, S1, S3; no S2 data, although S1 and S3 contain $Z \geq 55$ rows) against blind S2 MAPE for every fitted candidate in `results/model_comparison.csv`. The pocket formula is omitted because its V1 fit

diverged in the frozen run; that divergence depends on the optimizer path (§3.4), and with the converged V1 its selection score would be 4.19. The dashed line is Slater's rules (0 parameters).

The blind fit has three clear failures, which we report and have not corrected, since a correction now would be post hoc. Heavy p-block neutrals come out far too low (`results/known_failures.json`, refit on $Z \leq 54$): Pb at 1.20 eV against 7.42 eV, Tl at 1.74 against 6.11 and Rn at 6.10 against 10.75. The frozen run gave 1.19 and 1.73; refits in the two software environments differ here by about 0.01 eV. Lr comes out at -3.33 eV: the fitted r_c are negative and the relativistic bracket is not bounded, so the Z_{eff} bound alone does not keep the IE positive once $Z\alpha$ is large. Third IEs of lanthanides and actinides (4f/5f removal) are about 2.0–2.5 times too high (mean 2.2, median 2.1, 23 rows; worst Er^{2+} at 55.65 eV against 22.7 eV), because the $Z \leq 54$ training set contains no f electrons.

4.4 Hydrogen-like ions: correction layers

Table 3. 110 H-like ions, $Z = 1\text{--}110$ (`docs/first_principles.md` §2.1).

layer	MAPE (%)	median (%)	max (%)
Bohr Ry Z^2	6.0	4.2	19.5
(a) Dirac, point nucleus	0.190	0.118	0.91
(b) + recoil + finite nuclear size	0.113	0.109	0.248
(c) + one-loop QED (Uehling computed; F_{SE} from [31])	0.00117	0.000154	0.0096
(d) as (c) but with the closed-form $Z\alpha$ expansion of F_{SE}	0.354	0.191	1.18

The NIST H-like reference values are themselves computed from the same QED theory, so layer (c) shows consistency with that source rather than independent validation. The remaining $10^{-5}\text{--}10^{-4}$ residual at high Z is the omitted two-loop QED, nuclear-polarisation and recoil-QED terms.

4.5 Head-to-head with published screened hydrogenic models

We could specify two published SHMs fully from articles we were able to read: the parameter-free Kregar/Di Rocco model [20], [22], implemented from its definitions, and the relativistic model of Mendoza et al. [16], implemented from its published 19×19 table of fitted screening constants. The constants of More [11] and of Faussurier et al. [12] are in papers we could not access, and we found no verifiable reprint of their tables. We did not reconstruct them from memory.

4.5.1 Kregar/Di Rocco SHM (parameter-free; all 5847 rows)

Our implementation follows the published definitions: screening from hydrogenic densities with an exchange correction, iterated to self-consistency; energies $E = -\sum q_i Z_i^2 / 2n_i^2$; and non-relativistic, Pauli and Dirac variants. Its same-shell $Z \rightarrow \infty$ screening constants reproduce every printed digit (e.g. 1s 0.3125, 2p 0.3492), and its total energies agree with the published table within 0.71 %. Its cross-shell constants differ by 0.011 on average (at most 0.053), because the original uses fitted closed forms whose coefficients we could not obtain [19]. Valence IEs of near-neutral ions come out 3–5 eV higher than the printed model values; for Ar I we get 18.96 eV against 14.72 eV printed. Our near-neutral numbers therefore describe the model as defined, not the authors' code.

Table 4. Same 5847 rows, same scorer, MAPE in % (`results/lit_comparison.md`). For the 0-parameter models the S2 column is the error on the $Z \geq 55$ rows; for the fitted models it is the blind S2 refit.

model	params	all (median)	neutral 1st IE	H-like	charge ≥ 3	S2 ($Z \geq 55$)
Kregar/Di Rocco SHM, non-relativistic	0	13.6 (2.34)	230	6	–	15.1
Kregar/Di Rocco SHM + Pauli	0	12.6 (1.79)	225	1.16	–	13.9
Kregar/Di Rocco SHM + Dirac	0	12.5 (1.71)	224	0.19	4.68	13.6
Slater 1930	0	11.8 (7.48)	51.8	6	10.2	11.6
pocket formula	8	4.68 (3.04)	16.8	1.16	–	11.3
final	33	1.87 (0.94)	7.52	0.0012	1.61	6.6

Because of the valence discrepancy above, the 224 % neutral-atom figure probably overstates the error of the authors' own model on neutral atoms. The parameter-free SHM works well for ionised species, with a median of 1.54 % for

charge ≥ 3 . It fails near neutrality: 37 of its 5847 predictions are zero or negative, all for ions of charge 0–4 with an open d or f shell. Where the $Z \rightarrow \infty$ limit dominates, parameter-free and exact- σ_1 models both work. Near neutrality every parameter-free hydrogenic model degrades, and the fitted remainder is what makes the formula useful there, at the price of 8–33 parameters. The bounded fitted formula extrapolates better (6.6 %) than the parameter-free SHM scores on the same $Z \geq 55$ rows (13.6 %), while the unbounded fitted models extrapolate worse (57 % and 170 %).

4.5.2 Mendoza et al. 2011 (published constants; 5011 covered rows)

We transcribed the relativistic nlj screening constants of [16] (the 19×19 matrix $\sigma_{kk'}$, subshells $1s\frac{1}{2}$ to $5p\frac{3}{2}$) from Tables 1 and 2 of the authors' open-access deposit of the article (<https://oa.upm.es/11165/>), and implemented the model as published: Dirac energies of screened charges, $Q_k = Z - \sum_{k'} \sigma_{kk'}(P_{k'} - \delta_{kk'})$, and $IE = E_T(N-1) - E_T(N)$ with NIST ground configurations. The implementation reproduces six of the paper's printed tables to their rounding, the 84 IEs of its Table 3 to ≤ 0.023 % and its Tables 4–8 to ≤ 0.16 % (`models/benchmarks/mendoza2011/`). We fitted no parameter.

Four caveats apply. The authors fitted the constants with a genetic algorithm to NIST and FAC energies of isoelectronic sequences from He to Eu, with Z up to 92, which overlaps our test rows, so no split is held out for this model. Neutral atoms and singly charged ions were excluded from their fit. The constants were optimised for ionization and excitation energies together, so an IE-only score is not what they were optimised for. And the tables stop at $5p\frac{3}{2}$, so 836 rows ($Z \geq 55$, ground configurations with 5d, 5f, 6s, 6p, 6d or 7s electrons) have no prediction. The comparison is therefore restricted to the 5011 covered rows.

Table 4a. The 5011 rows covered by Mendoza et al.; same scorer, all three models fitted to all data (in-sample). Cells: mean / median absolute percentage error [n]. Source: `tools/compare_mendoza.py` $\rightarrow$ `results/compare_mendoza.md`.

rows	Screened Rydberg (final, 33 p)	bounded 9-parameter (pa_bound9)	Mendoza et al. 2011
all covered rows	1.62 / 0.78 [5011]	2.63 / 1.19	2.82 / 0.85
ions only ($Z > N$)	1.55 / 0.76 [4957]	2.54 / 1.17	2.56 / 0.82
charge ≥ 3	1.45 / 0.73 [4837]	2.43 / 1.14	2.27 / 0.77
$N \leq 10$	0.44 / 0.23 [1048]	0.51 / 0.30	0.40 / 0.17
$11 \leq N \leq 36$	1.43 / 0.82 [2275]	2.12 / 1.17	1.82 / 0.61
$N \geq 37$	2.62 / 1.51 [1688]	4.65 / 3.52	5.68 / 2.48
$Z \geq 55$	1.57 / 0.72 [3526]	2.30 / 1.08	2.74 / 1.00
H-like	0.0012 [110]	0.0012	0.19
experimental	4.37 [236]	7.17	11.3
neutral atoms (outside Mendoza's fit range)	8.37 [54]	11.5	27.1
99th percentile / max APE	11.3 / 30.4	14.8 / 38.0	28.4 / 119
rows within 5 %	92.2 %	83.8 %	86.4 %

On the ions it covers, the 33-parameter formula is competitive with the published constants of Mendoza et al., not decisively better. Its mean error is lower (1.62 % vs 2.82 %), its median is similar (0.78 % vs 0.85 %), and it has far fewer large misses (99th percentile 11 % vs 28 %). Mendoza et al. are more accurate for few-electron ions ($N \leq 10$: 0.40 % vs 0.44 %) and have the lower median for $11 \leq N \leq 36$. Row by row, the formula is closer to NIST on 51.3 % of the covered rows. The 9-parameter variant matches their mean on ions (2.54 % vs 2.56 %) with a larger median.

The two models differ in size and scope. Mendoza et al. publish a 19×19 matrix with 331 non-zero constants, fitted by the authors; we count these as published numbers rather than independent degrees of freedom, and they are the only model-specific numbers our implementation of their IE prescription uses. The formula has 33 global parameters and was also validated on held-out splits. Their model also yields excitation energies and orbital properties and is used in plasma codes, while ours gives only ionization energies. The neutral-atom row lies outside the range they fitted and is shown for completeness. The experimental-row value here (236 rows) differs from the all-row value of §3.1 (4.63 % on 311 rows) only because the row sets differ.

A comparison with Dirac–Fock ionization energies [30] on the same rows is left for future work.

4.6 The exact first-order screening constants compared with Slater's

Table 5. σ_1 for neutral-atom ground configurations, compared with Slater's σ . Excerpt; the full table for $N = 1$ –110 is in `results/fp_zexp_coefficients.csv`, with values for all 5847 rows in `results/fp_zexp_rows_coefficients.csv`.

N	configuration	removed	E_1 (exact)	ΔE_1	σ_1 (ab initio)	σ (Slater)
2	$1s^2$	1s	5/8	−0.625000	0.6250	0.30
3	$1s^2 2s$	2s	5965/5832	−0.397805	1.5912	1.70
4	$1s^2 2s^2$	2s	586373/373248	−0.536469	2.1459	2.05
6	$1s^2 2s^2 2p^2$ (3P)	2p	10957679/3359232	−0.931338	3.7254	2.75
8	$1s^2 2s^2 2p^4$	2p	4754911/839808	−1.308370	5.2335	3.45
10	$1s^2 2s^2 2p^6$	2p	2455271/279936	−1.636495	6.5460	4.15
11	[Ne]3s	3s	9.635901	−0.865071	7.7856	8.80
18	[Ne]3s ² 3p ⁶	3p	17.980333	−1.407535	12.6678	11.25
19	[Ar]4s	4s	18.859971	−0.879639	14.0742	16.80
29	[Ar]3d ¹⁰ 4s	4s	39.317547	−1.347129	21.5541	25.30
36	[Ar]3d ¹⁰ 4s ² 4p ⁶	4p	50.145538	−1.667284	26.6765	27.75

E_1 is the Hund-term single-configuration value, except for $N = 4$ (Be), where σ_1 uses the Layzer-complex value 1.5592742. Decimal E_1 entries are exact rationals stored in the project cache, and $\sigma_1 = -n^2 \Delta E_1$.

Slater's rules describe the neutral atom, while σ_1 is the exact screening in the bare-nucleus limit, and the two disagree in a systematic way (Figure 3). For inner electrons the hydrogenic same-shell screening is stronger than Slater's 0.35 per electron (Ne 2p: 6.55 against 4.15). For valence electrons of heavy atoms σ_1 is much smaller than Slater's σ and than the screening implied by the measured IE (Na 3s: 7.79 against Slater 8.80 and an empirical 9.16; Cs 6s: 42.8 against Slater 52.8). The fitted remainder D supplies this difference near neutrality.

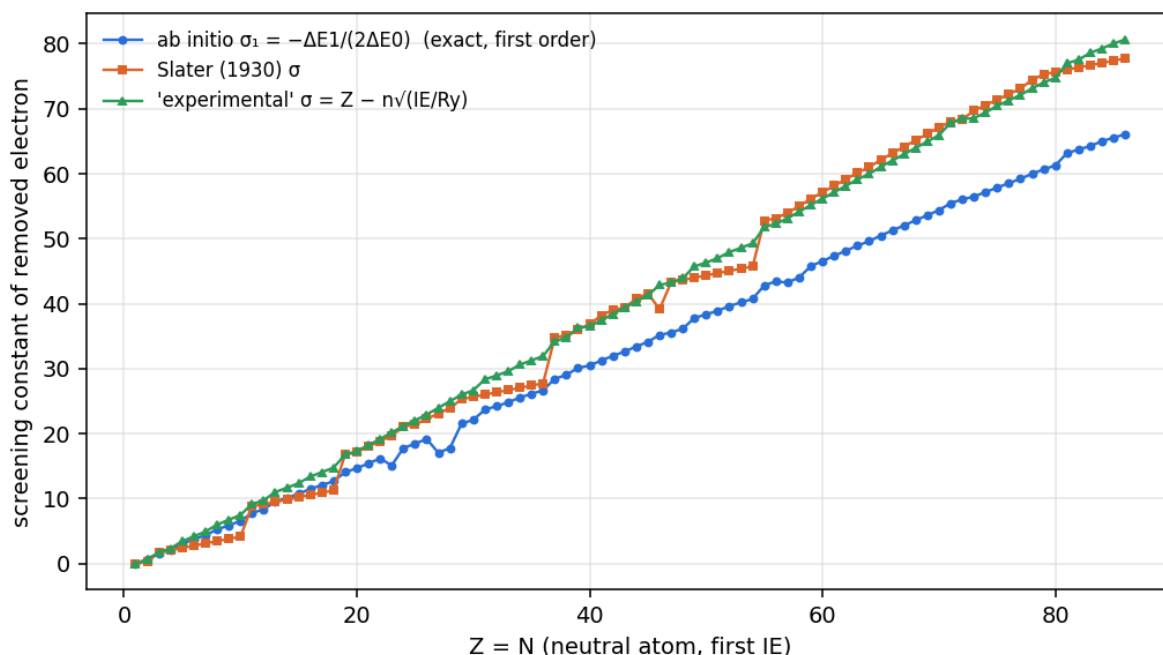


Figure 3. Exact first-order screening σ_1 against Slater's σ for neutral-atom configurations.

4.7 Worked examples and coverage

For the first IE of oxygen ($1s^2 2s^2 2p^4 \rightarrow 2p^3$, NIST 13.618 eV), $\sigma_1 = 5.23348$ and $T = 3.3496$. With $h = 1.76652$ this gives $D = 0.71285$ and $Z_{\text{eff}} = 2.05367$, which lies between $Z_a = 1$ and $Z = 8$. The hydrogenic term is 14.3457 eV, or 14.3162 eV after relativity, and the Hund term is −0.8365 eV, so IE = 13.479 eV, an error of −1.02 %.

Nitrogen ($1s^2 2s^2 2p^3 \rightarrow 2p^2$, NIST 14.534 eV) has a half-filled shell. Here $\sigma_1 = 4.37867$ and $T = 2.9470$; with $h = 1.62133$, $D = 0.63782$ and $Z_{\text{eff}} = 1.98351$. The hydrogenic term is 13.3822 eV, or 13.3626 eV after relativity. The Hund term is +0.8365 eV, since $K_p(3) = +1.2$ has the opposite sign to oxygen's, and $IE = 14.199$ eV, an error of -2.31% .

For Mg^{2+} (NIST 80.144 eV) the formula gives 78.63 eV (-1.88%).

Table 6. Case studies: first IEs of six neutral atoms, with every intermediate quantity, for the three named models. Printed by `tools/audit_components.py` and `tools/case_study_table.py` (`results/case_studies.md`); the script asserts that the components reproduce the production code to 10^{-9} . D is the screening remainder. The relativistic factor is $F_{\{n,j\}} \cdot R$ for the σ_1 models ($R = 1$ for `pa_bound9`) and the Sommerfeld bracket for the pocket formula.

atom (removed)	model	σ_1	v_g (same, in, core, df, out)	T	h	D	Z_{eff}	Ry Z_{eff}^2/n^2 (eV)	rel. factor	Hund (eV)	IE (eV)	NIST (eV)	error
O (2p)	final	5.2335	3, 4, 0, 0, 0	3.3496	1.7665	0.7128	2.0537	14.3457	0.99794	-0.8365	13.479	13.618	-1.02 %
O (2p)	pa_bound9	5.2335	3, 4, 0, 0, 0	3.0925	1.7665	0.6173	2.1492	15.7116	1.00002	-1.4702	14.241	13.618	+4.58 %
O (2p)	pocket	-	3, 4, 0, 0, 0	-	-	6.1394*	1.8606	11.7747	1.00001	-2.0735	9.701	13.618	-28.76 %
N (2p)	final	4.3787	2, 4, 0, 0, 0	2.9470	1.6213	0.6378	1.9835	13.3822	0.99853	+0.8365	14.199	14.534	-2.31 %
N (2p)	pa_bound9	4.3787	2, 4, 0, 0, 0	2.8578	1.6213	0.5691	2.0522	14.3258	1.00001	+1.4702	15.796	14.534	+8.68 %
N (2p)	pocket	-	2, 4, 0, 0, 0	-	-	5.2632*	1.7368	10.2598	1.00001	+2.0735	12.333	14.534	-15.14 %
Na (3s)	final	7.7856	0, 8, 2, 0, 0	7.1802	2.2144	1.1877	2.0266	6.2091	0.99905	0	6.203	5.139	+20.70 %
Na (3s)	pa_bound9	7.7856	0, 8, 2, 0, 0	10.6892	2.2144	1.3219	1.8925	5.4142	1.00005	0	5.414	5.139	+5.36 %
Na (3s)	pocket	-	0, 8, 2, 0, 0	-	-	8.9608*	2.0392	6.2862	1.00006	0	6.287	5.139	+22.33 %
Ca (4s)	final	14.6706	1, 8, 10, 0, 0	24.4604	4.3294	2.8939	2.4356	5.0444	0.99659	0	5.027	6.113	-17.77 %
Ca (4s)	pa_bound9	14.6706	1, 8, 10, 0, 0	34.5745	4.3294	3.0747	2.2548	4.3233	1.00005	0	4.323	6.113	-29.28 %
Ca (4s)	pocket	-	1, 8, 10, 0, 0	-	-	17.5935*	2.4065	4.9248	1.00006	0	4.925	6.113	-19.43 %
Fe (4s)	final	19.1852	1, 14, 10, 0, 0	30.9931	5.8148	3.8109	3.0039	7.6732	0.99190	0	7.611	7.902	-3.69 %
Fe (4s)	pa_bound9	19.1852	1, 14, 10, 0, 0	38.1569	5.8148	3.8852	2.9296	7.2983	1.00009	0	7.299	7.902	-7.64 %
Fe (4s)	pocket	-	1, 14, 10, 0, 0	-	-	22.8597*	3.1403	8.3855	1.00011	0	8.386	7.902	+6.12 %
Pb (6p)	final	63.7240	1, 20, 60, 0, 0	153.9342	17.2760	13.1422	5.1339	9.9612	0.79963	+0.0465	8.012	7.417	+8.02 %
Pb (6p)	pa_bound9	63.7240	1, 20, 60, 0, 0	189.5551	17.2760	13.3197	4.9564	9.2843	1.00019	+0.0817	9.368	7.417	+26.31 %
Pb (6p)	pocket	-	1, 20, 60, 0, 0	-	-	76.6038*	5.3962	11.0052	1.00023	+0.1152	11.123	7.417	+49.97 %

*Pocket formula: total screening $\Sigma s_g v_g + t(N-1)/(Z_a + \kappa)$; it has no σ_1 and no bound.

For s removal (Na, Ca, Fe) the IE is the screened Rydberg term times relativistic factors within about 1 % of unity; no model in this paper adds an s-type correction. The final model is the most accurate of the three on O, N, Ca, Fe and Pb, pa_bound9 only on Na, and the pocket formula on none. Both bounded models fail on Ca (−17.8 % and −29.3 %). For Pb, $\sigma_1 = 63.72$ (Figure 3), while the total screening $\sigma_1 + D$ is 76.87.

We evaluated the public API for every $Z = 1-118$ and every $N = 1-Z$, 7021 values. All are finite and positive, with no crashes. There are 22 violations of the monotonicity $IE(Z, N-1) > IE(Z, N)$: 6 at Pt–Bi, caused by 4f/5s ordering, and 16 at Rf–Ds, where the input ground configuration jumps between neighbouring ions (e.g. Rf $N = 68$ is $4f^{12}6s^2$ and $N = 69$ is $4f^{14}5d^1$).

Above $Z = 110$ the predictions are qualitative. The 7p neutrals come out too low: Og is predicted at 3.66 eV, far below its lighter congener Rn (NIST 10.75 eV), again because of the negative r_c .

5. Discussion

5.1 What is new and what is rediscovered

Much of the formula is known. The $1/Z$ expansion and its exact first-order term are Layzer's (σ_1 is his $Z \rightarrow \infty$ screening constant [4]), computed with textbook Slater-integral algebra [33], [34]. The screened hydrogenic form goes back to [1], [10], [11], and self-consistent (Z, N)-dependent screening to [17], [20]. The $1/(Z_a + \kappa)$ remainder is Edlén-type isoelectronic behaviour [8]. The one-electron Dirac, recoil, finite-size and QED corrections come from [31], [32], and LSDA Δ SCF from [27], [28].

Four things may be new. Z-expansion work [4], [5], [9] computed first-order energies for selected configurations and isoelectronic sequences, and we are not aware of a table of exact first-order screening constants for every NIST ground configuration with $N = 1-110$; we offer ours as a complete tabulation, not as a new quantity. The formula joins exact σ_1 to a bounded fitted remainder in one closed form and is validated on all 5847 NIST successive IEs, with a pre-registered selection score and an extrapolation from $Z \leq 54$ to $Z \geq 55$; we found no out-of-range extrapolation test in the SHM literature we read. Two published SHMs are scored on the same rows: the Kregar/Di Rocco model on all 5847, and the constants of Mendoza et al. on the 5011 they cover, where the formula is competitive rather than decisively better (§4.5). Finally, the 8-parameter pocket formula is a hand-calculable successor to Slater's rules, with 4.68 % against 11.8 % on all rows and 11.3 % against 11.6 % on blind S2.

We do not claim a first formula for all ionization energies, since SHMs and Dirac–Fock tables already cover all ions, or a first-principles formula, since only σ_1 and the one-electron corrections are first principles and the final model has 33 fitted parameters. We did not search the machine-learning literature on ionization energies systematically and make no claim relative to it.

Fitting alone interpolates to about 1.6 % and extrapolates catastrophically (170 %). Exact theory alone extrapolates smoothly but is about 70 % off overall. Fixing the Z^2 and Z terms by theory and confining the fit to a bounded $O(Z^0)$ remainder gives about 2 % inside the data and 6.6 % in blind extrapolation.

5.2 Where the formula fails

Heavy near-neutral atoms, alkaline earths and noble gases are the main failure. The blind neutral first-IE MAPE is 21.8 %, and the all-data neutral MAPE is 7.5 % (6.4 % for u35). Even with all data fitted (`results/known_failures.json`) the ns^2 alkaline earths are too low (Ca −17.8 %, Sr −14.8 %, Ba −8.4 %), Rn is too low by 24.6 %, and Na is too high by 20.7 % (Table 6). Pb's small all-data error (+8.0 %) is partly a cancellation. Its Rydberg term alone is 9.96 eV (+34 % against NIST 7.42 eV); the relativistic bracket, 0.80 because $r_{p\frac{1}{2}}$ is negative, lowers it to 7.97 eV (+7.4 %), and the final IE is 8.01 eV (Table 6; `results/audit_components.md`). A term of the wrong physical sign is cancelling an overestimate. Valence screening in a neutral atom is an all-order, non-perturbative effect, and the parameter-free LSDA Δ SCF does better on neutral atoms (3.3 % for $Z \leq 54$).

The relativistic term is unbounded. The fitted r_c are all negative (−0.45 to −1.47), opposite to the direct relativistic contraction of s and $p_{\frac{1}{2}}$ electrons. Because the bracket is not bounded, the $Z \leq 54$ fit gives a negative Lr and low

superheavy 7p IEs. A bounded relativistic term chosen on V1/V2/S1/S3 alone is the obvious next step; the first attempt (§3.3) fixed the signs but lowered the selection score.

f-electron removal is the least accurate class, with 3.5 % on all data and third IEs about 2.0–2.5 times too high (mean 2.2) in blind S2.

The formula has one Hund term per removed electron and no term-dependent multiplet energies, and ions whose ground configuration rearranges on ionization are described by single-configuration inputs. These limits also cause the 22 monotonicity violations.

19 of the 33 parameters are shrunk class deviations. The bounded 9-parameter variant pa_bound9 (selection 2.51, blind S2 7.80 %, all data 2.90 %; parameters in Appendix A) is a reasonable alternative with fewer parameters, though it was not the pre-registered winner.

Most NIST values are labelled theoretical or semi-empirical (4617 + 919 of 5847), so agreement with them is partly agreement with other theory.

5.3 Why no exact closed form exists for $N \geq 2$

With $V = \sum 1/r_{ij}$ the many-electron Schrödinger equation does not separate. $E(Z)$ is analytic in $1/Z$ only up to a critical charge; for He, $1/Z_c \approx 1.0975$. Beyond first order every coefficient E_k ($k \geq 2$) is an infinite sum over the hydrogenic continuum, with no known closed form even for two electrons, and is known only numerically [6]. A closed formula for all ions must therefore combine exact low orders with an approximation for the rest. The choice is which approximation, and how it is validated. We keep the exact orders exact and confine the fit to the remainder, under a bound that prevents it from extrapolating wildly.

6. Conclusions

The screened Rydberg formula gives every successive ionization energy of every element from the configuration alone, with 33 global parameters. On 5847 NIST values its MAPE is 1.87 % (median 0.94 %), 7.5 % for neutral atoms and 0.0012 % for H-like ions. Fitted on $Z \leq 54$, it predicts $Z = 55$ –110 with 6.60 % MAPE, below the 11.6 % of Slater's rules and the 13.6 % of a published parameter-free screened hydrogenic model, because first-order perturbation theory fixes the large- Z behaviour exactly and the fitted remainder is bounded.

The formula still fails in known places. In blind S2, Pb comes out at about 1.2 eV against 7.42 and Tl at about 1.7 eV against 6.11. Lr is negative in the $Z \leq 54$ fit, and Og (3.66 eV) lies far below Rn. Alkaline-earth neutrals are too low, f-electron removal is too high in extrapolation, and all fitted relativistic coefficients are negative.

The most reusable result is the table of exact rational first-order screening constants σ_1 for $N = 1$ –110.

Data and code availability

All code, data and results are in the project repository, <https://github.com/AwaisSDev/screened-rydberg-formula>, archived on Zenodo as version 1.0.0 (doi:10.5281/zenodo.23211017). It contains the NIST table (data/nist_ie.csv) and the shared scorer (evaluate.py); the exact first-order coefficients (results/fp_zexp_coefficients.csv, results/fp_zexp_rows_coefficients.csv); the final model and its parameters (models/push_a/model.py, models/unified/final.py, results/uni_final_params.json); predictions for all rows (results/uni_predictions.csv); the validation script (models/unified/validate_blind.py); the literature re-implementations (models/literature/kregar_shm.py, models/benchmarks/mendoza2011/); every per-row model input (results/model_inputs.csv) with the stand-alone reference implementation (tools/verify_from_inputs.py), which reproduces the production predictions to $7 \cdot 10^{-16}$; and the component audit (tools/audit_components.py, Table 6) and benchmark matrix with row counts in every cell (tools/benchmark_matrix.py → results/benchmark_matrix.md). A command-line and Python interface (ionization.py) evaluates the formula for any $Z \leq 118$ and $N \leq Z$.

AI-assistance disclosure

The computations, code, analysis and a draft of this text were produced with AI agents (Anthropic Claude) working under the author's direction. The author designed and directed the study and is responsible for the content. Every number in the paper is generated by a script in the repository and can be regenerated from it. Bibliographic data were checked against Crossref. No AI system is listed as an author.

Acknowledgements

None.

References

- [1] J. C. Slater, "Atomic shielding constants," *Phys. Rev.*, vol. 36, pp. 57–64, 1930.
- [2] E. Clementi and D. L. Raimondi, "Atomic screening constants from SCF functions," *J. Chem. Phys.*, vol. 38, pp. 2686–2689, 1963.
- [3] E. Clementi, D. L. Raimondi, and W. P. Reinhardt, "Atomic screening constants from SCF functions. II. Atoms with 37 to 86 electrons," *J. Chem. Phys.*, vol. 47, pp. 1300–1307, 1967.
- [4] D. Layzer, "On a screening theory of atomic spectra," *Ann. Phys. (N.Y.)*, vol. 8, pp. 271–296, 1959, doi: 10.1016/0003-4916(59)90023-5.
- [5] D. Layzer, Z. Horák, M. N. Lewis, and D. P. Thompson, "Second-order Z-dependent theory of many-electron atoms," *Ann. Phys. (N.Y.)*, vol. 29, no. 1, pp. 101–124, 1964, doi: 10.1016/0003-4916(64)90192-7.
- [6] C. W. Scherr and R. E. Knight, "Two-electron atoms III. A sixth-order perturbation study of the 1^1S ground state," *Rev. Mod. Phys.*, vol. 35, no. 3, pp. 436–442, 1963, doi: 10.1103/RevModPhys.35.436.
- [7] A. Dalgarno and A. L. Stewart, "A perturbation calculation of properties of the helium iso-electronic sequence," *Proc. R. Soc. Lond. A*, vol. 247, no. 1249, pp. 245–259, 1958, doi: 10.1098/rspa.1958.0182.
- [8] B. Edlén, "Atomic spectra," in *Handbuch der Physik / Encyclopedia of Physics*, vol. 27, *Spectroscopy I*, S. Flügge, Ed. Berlin, Germany: Springer, 1964, pp. 80–220, doi: 10.1007/978-3-662-35391-2_2.
- [9] U. I. Safronova, I. Yu. Tolstikhina, R. Bruch, T. Tanaka, F. Hao, and D. Schneider, "Screening theory for transition energies of highly charged ions," *Phys. Scr.*, vol. 47, no. 3, pp. 364–382, 1993, doi: 10.1088/0031-8949/47/3/007.
- [10] H. Mayer, "Methods of opacity calculations," Los Alamos Sci. Lab., Los Alamos, NM, USA, Rep. LA-647, 1947.
- [11] R. M. More, "Electronic energy levels in dense plasmas," *J. Quant. Spectrosc. Radiat. Transf.*, vol. 27, no. 3, pp. 345–357, 1982, doi: 10.1016/0022-4073(82)90127-3.
- [12] G. Faussurier, C. Blancard, and A. Decoster, "New screening coefficients for the hydrogenic ion model including l-splitting for fast calculations of atomic structure in plasmas," *J. Quant. Spectrosc. Radiat. Transf.*, vol. 58, no. 2, pp. 233–260, 1997, doi: 10.1016/S0022-4073(97)00018-6.
- [13] G. Faussurier, C. Blancard, and P. Renaudin, "Equation of state of dense plasmas using a screened-hydrogenic model with l-splitting," *High Energy Density Phys.*, vol. 4, pp. 114–123, 2008.
- [14] P. Martel, J. G. Rubiano, J. M. Gil, L. Doreste, and E. Mínguez, "Analytical expressions for the n-order momenta of charge distribution for ions," *J. Quant. Spectrosc. Radiat. Transf.*, vol. 60, no. 4, pp. 623–633, 1998, doi: 10.1016/S0022-4073(97)00226-4.
- [15] J. G. Rubiano, R. Rodríguez, J. M. Gil, F. H. Ruano, P. Martel, and E. Mínguez, "A screened hydrogenic model using analytical potentials," *J. Quant. Spectrosc. Radiat. Transf.*, vol. 72, no. 5, pp. 575–588, 2002, doi: 10.1016/S0022-4073(01)00142-X.
- [16] M. A. Mendoza, J. G. Rubiano, J. M. Gil, R. Rodríguez, R. Florido, P. Martel, and E. Mínguez, "A new set of relativistic screening constants for the screened hydrogenic model," *High Energy Density Phys.*, vol. 7, no. 3, pp. 169–179, 2011,

doi: 10.1016/j.hedp.2011.04.006.

- [17] M. Kregar, “The virial and the independent particle models of the atom,” *Phys. Scr.*, vol. 29, no. 5, pp. 438–447, 1984, doi: 10.1088/0031-8949/29/5/005.
- [18] M. Kregar, “The virial as the atomic model potential energy operator,” *Phys. Scr.*, vol. 31, no. 4, pp. 246–254, 1985, doi: 10.1088/0031-8949/31/4/005.
- [19] H. O. Di Rocco, *Braz. J. Phys.*, vol. 22, p. 227, 1992 (as cited in [20]; not independently verified).
- [20] J. Pomarico, D. I. Iriarte, and H. O. Di Rocco, “An efficient screening approach to be used in plasma modeling and ion-surface collision experiments,” *Braz. J. Phys.*, vol. 35, no. 1, pp. 130–135, 2005, doi: 10.1590/S0103-97332005000100008.
- [21] F. Lanzini and H. O. Di Rocco, “Screening parameters for the relativistic hydrogenic model,” *High Energy Density Phys.*, vol. 17, pp. 240–247, 2015, doi: 10.1016/j.hedp.2015.08.002.
- [22] H. O. Di Rocco and F. Lanzini, “Breit and quantum electrodynamics energy contributions in multielectron atoms from the relativistic screened hydrogenic model,” *Braz. J. Phys.*, vol. 46, pp. 175–183, 2016, doi: 10.1007/s13538-015-0397-9.
- [23] B. F. Rozsnyai, “Relativistic Hartree-Fock-Slater calculations for arbitrary temperature and matter density,” *Phys. Rev. A*, vol. 5, no. 3, pp. 1137–1149, 1972, doi: 10.1103/PhysRevA.5.1137.
- [24] H.-K. Chung, M. H. Chen, W. L. Morgan, Yu. Ralchenko, and R. W. Lee, “FLYCHK: Generalized population kinetics and spectral model for rapid spectroscopic analysis for all elements,” *High Energy Density Phys.*, vol. 1, pp. 3–12, 2005.
- [25] A. J. Crilly *et al.*, “SpK: A fast atomic and microphysics code for the high-energy-density regime,” *High Energy Density Phys.*, 2023, doi: 10.1016/j.hedp.2023.101053.
- [26] T. Koopmans, “Über die Zuordnung von Wellenfunktionen und Eigenwerten zu den einzelnen Elektronen eines Atoms,” *Physica*, vol. 1, pp. 104–113, 1934.
- [27] W. Kohn and L. J. Sham, “Self-consistent equations including exchange and correlation effects,” *Phys. Rev.*, vol. 140, pp. A1133–A1138, 1965.
- [28] S. Kotochigova, Z. H. Levine, E. L. Shirley, M. D. Stiles, and C. W. Clark, “Local-density-functional calculations of the energy of atoms,” *Phys. Rev. A*, vol. 55, pp. 191–199, 1997.
- [29] S. J. Chakravorty, S. R. Gwaltney, E. R. Davidson, F. A. Parpia, and C. Froese Fischer, “Ground-state correlation energies for atomic ions with 3 to 18 electrons,” *Phys. Rev. A*, vol. 47, pp. 3649–3670, 1993.
- [30] G. C. Rodrigues, P. Indelicato, J. P. Santos, P. Patti, and F. Parente, “Systematic calculation of total atomic energies of ground state configurations,” *At. Data Nucl. Data Tables*, vol. 86, pp. 117–233, 2004, doi: 10.1016/j.adt.2003.11.005.
- [31] V. A. Yerokhin and V. M. Shabaev, “Lamb shift of $n = 1$ and $n = 2$ states of hydrogen-like atoms, $1 \leq Z \leq 110$,” *J. Phys. Chem. Ref. Data*, vol. 44, 033103, 2015.
- [32] A. Kramida, Yu. Ralchenko, J. Reader, and NIST ASD Team, *NIST Atomic Spectra Database* (version 5.12). Gaithersburg, MD, USA: Nat. Inst. Standards Technol., 2024. [Online]. Available: <https://physics.nist.gov/asd> (accessed on or before Oct. 5, 2026), doi: 10.18434/T4W30F.
- [33] R. D. Cowan, *The Theory of Atomic Structure and Spectra*. Berkeley, CA, USA: Univ. California Press, 1981.
- [34] C. Froese Fischer, T. Brage, and P. Jönsson, *Computational Atomic Structure: An MCHF Approach*. Bristol, U.K.: Inst. Phys. Publ., 1997.

Appendix A. Fitted parameters (all-data fit)

Table A1. Source: `results/uni_final_params.json`, which is identical to `results/pa_params.json`.

block	values
τ_g (same, in, core, df, out)	0.3928, 0.7879, 2.2243, 2.5488, 10.5688
κ	1.7527 (1.8027 as used in the code, which applies $ \kappa + 0.05$)
$\delta\tau_c$ (19 classes, defined in Table A2)	same_s -0.1552, same_p 0.0098, same_d 0.0004, same_f 0.1457, sn_in_p -0.1133, n1_sp_sp -0.3916, n1_sp_d 0.3009, n1_sp_f 0.2042, n2_sp_sp -0.2195, n2_sp_df -0.0622, d_near -0.6962, n1_d_d -0.2689, n1_d_f 0.3286, f_near 0.5476, n1_f_f -0.0918, out_d -0.0176, out_f 0.0173, core_sp 0.2828, core_df 0.1812
r_c (s, p $\frac{1}{2}$, p $\frac{3}{2}$, d, f)	-0.4543, -0.8129, -1.1533, -0.8426, -1.4672
x_l (p, d, f)	0.2049, 0.4402, 0.4612

Bounded 9-parameter variant, pa_bound9 (all-data fit, results/pa_params.json, candidate pa_bound9). Same equation (2.2) and inputs as the final model, with $T = \sum_g \tau_g v_g$ (no class deviations), $R = 1$ (no relativistic bracket), and the same $F_{\{n,j\}}$, μ and QED/FNS terms. Using these 4-decimal values instead of full precision changes no IE by more than 0.002 % (tools/hand_recompute.py).

block	values
τ_g (same, in, core, df, out)	0.2347, 0.5971, 2.9563, 2.9515, 23.7087
κ	2.2091 (2.2591 as used)
x_l (p, d, f)	0.3602, 0.4237, 0.8215

Metrics: all data 2.90 % (median 1.39 %), neutral 12.0 %, H-like 0.0012 %; V1 2.16, V2 1.95, S1 2.90, S3 3.05, selection score 2.51; blind S2 7.80 % (Table 1).

Table A2. Screening classes c and groups g of §2.2. (n, l) is the removed subshell and (n', l') the subshell of another electron. Each class lies inside one group. "rows" is the number of the 5847 NIST rows with $v_c > 0$. $\delta\tau_c$ is from Table A1, and $\tau_g + \delta\tau_c$ is the coefficient of one electron of that class in T . (Verified by tools/check_grouping_rule.py.)

group g	class c	target l	other electron (n', l')	rows	$\delta\tau_c$	$\tau_g + \delta\tau_c$
same	same_s	s	$(n', l') = (n, l)$	534	-0.1552	0.2376
same	same_p	p	$(n', l') = (n, l)$	1713	0.0098	0.4026
same	same_d	d	$(n', l') = (n, l)$	1720	0.0004	0.3932
same	same_f	f	$(n', l') = (n, l)$	717	0.1457	0.5385
in	sn_in_p	p	$n' = n, l' = s$	2073	-0.1133	0.6745
in	n1_sp_sp	s, p	$n' = n - 1, l' = s, p$	2912	-0.3916	0.3963
in	n1_sp_d	s, p	$n' = n - 1, l' = d$	1248	0.3009	1.0888
in	n1_sp_f	s, p	$n' = n - 1, l' = f$	386	0.2042	0.9921
core	n2_sp_sp	s, p	$n' = n - 2, l' = s, p$	2083	-0.2195	2.0048
core	n2_sp_df	s, p	$n' = n - 2, l' = d, f$	668	-0.0622	2.1621
core	core_sp	s, p	$n' \leq n - 3$	1311	0.2828	2.5070
df	d_near	d	$n' = n$ or $n - 1, l' = s, p$	1938	-0.6962	1.8526
df	n1_d_d	d	$n' = n - 1, l' = d$	1078	-0.2689	2.2799
df	n1_d_f	d	$n' = n - 1, l' = f$	391	0.3286	2.8773
df	f_near	f	$n' = n$ or $n - 1, l' = s, p, d$	778	0.5476	3.0964
df	n1_f_f	f	$n' = n - 1, l' = f$	201	-0.0918	2.4569
df	core_df	d, f	$n' \leq n - 2$	2716	0.1812	2.7300
out	out_d	d	$n' > n$	15	-0.0176	10.5512
out	out_f	f	$n' > n$	294	0.0173	10.5861
out	sn_out	any	$n' = n, l' > l$	0	none	10.5688
out	out_sp	s, p	$n' > n$	0	none	10.5688

Pocket formula (8 parameters, all-data fit, results/uni_params.json). This is the only model called "pocket formula" in this paper.

$$Z_{\text{eff}} = Z - \sum_g s_g \nu_g - t \frac{N-1}{Z-N+1+\kappa}, \quad \text{IE} = \text{Ry} \frac{Z_{\text{eff}}^2}{n^2} \left[1 + \frac{(Z_{\text{eff}} \alpha)^2}{n^2} \left(\frac{n}{j + \frac{1}{2}} - \frac{3}{4} \right) \right] + \text{Ry} \frac{x K_l(k)}{n^2}$$

The v_g are the five group counts of §2.2. The fitted values are s_{same} 0.7499, s_{in} 0.7514, s_{core} 0.8432, s_{df} 0.8408, s_{out} 1.0497, t 1.3725, κ 9.815 (9.865 as used) and x 0.508.